Big Data | 6CS030

Interaction with MongoDB using `mongosh`

NoSQL databases

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1. MongoDB Workshop Exercises

1.1. Exercise 2.1

1.1.1. Inserting Departments

```
------

test> db.dept.insert(
| {
| _id: 10,
| deptno: 10,
| dname: "ACCOUNTING",
| loc: "NEW YORK"
| })
|
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
{ acknowledged: true, insertedIds: { '0': 10 } }
test> db.dept.insert(
| {
| _id:
| 20,
| deptno: 20,
| dname: "RESEARCH",
| loc:
| "DALLAS"
| })
{ acknowledged: true, insertedIds: { '0': 20 } }
test> db.dept.insert(
| {
| _id:
| 40,
| deptno: 40,
| dname: "OPERATIONS",
| loc:
| "BOSTON"
| })
```

Figure 1: Insert Departments

```
test> db.dept.insert(
| {
| _id:
| 10,
| deptno: 50,
| dname: "TEST",
| loc:
| "WOLVERHAMPTON"
| })
Uncaught:
MongoBulkWriteError: E11000 duplicate key error collection: test.dept index: _id_ dup key: { _id: 10 }
Result: BulkWriteResult {
  insertedCount: 0,
  matchedCount: 0,
  modifiedCount: 0,
  deletedCount: 0,
  upsertedCount: 0,
  upsertedIds: {},
  insertedIds: {}
}
Write Errors: [
  WriteError {
    err: {
      index: 0,
      code: 11000,
      errmsg: 'E11000 duplicate key error collection: test.dept index: _id_ dup key: { _id: 10 }',
      errInfo: undefined,
      op: { _id: 10, deptno: 50, dname: 'TEST', loc: 'WOLVERHAMPTON' }
    }
  }
]
test>
```

Figure 2: Duplicate Key write error

1.1.2. Question 01: comparing INSERT operation

1. **Compare how you added the above data and how it differs from INSERT records in a relational database**

Ans: We directly added the data into the db using `insert` command while in the relational db we had to first define the schema of table (how the data we are going to store looks like) and then only we could insert data according to that schema/interface defined.

2. **Add Department 30, which has the following `key/values: _id: 30, deptno: 30, dname: SALES and loc: CHICAGO`**

```
test> show collections
dept
test> db.dept.insert({
| _id: 30,
| deptno: 30,
| dname: "SALES",
| loc: "CHICAGO",
| })
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
{ acknowledged: true, insertedIds: { '0': 30 } }
test>
```

Figure 3: Inserting new department

1.2. Exercise 2.2

1.2.1. Inseting bulk employee

```
test> db.emp.insert( [
| {
| empno: 7782,
| ename: 'CLARK',
| job: 'MANAGER',
| mgr: 7839,
| hiredate: new Date('1989-06-09'),
| sal: 2450,
| deptno: 10
| },
| {
| empno:7839,
| ename: 'KING',
| job: 'PRESIDENT',
| hiredate: new Date('1980-11-17'),
| sal: 5000,
| deptno: 10
| },
| {
| empno: 7934,
| ename: 'MILLER',
| job: 'CLERK',
| mgr: 7782,
| hiredate: new Date('1985-01-23'),
| sal: 1300,
| deptno: 10
| }
| ]
| )
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69bca99a7205df8fc48563b1'),
    '1': ObjectId('69bca99a7205df8fc48563b2'),
    '2': ObjectId('69bca99a7205df8fc48563b3')
  }
}
```

Figure 4: Bulk Inserting Emp

1.2.2. Question 01: adding table employees

Add the employees for Department 30. Remember this time, some employees will have a commission (COMM):

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7499	ALLEN	SALES-MAN	7698	20-FEB-95	1600	300	30
7698	BLAKE	MANAGER	7839	01-MAY-81	2850		30
7900	JAMES	CLERK	7698	03-DEC-81	950		30
7654	MARTIN	SALES-MAN	7698	28-SEP-93	1250	1400	30
7844	TURNER	SALES-MAN	7698	08-SEP-81	1500	0	30
7521	WARD	SALES-MAN	7698	22-FEB-94	1250	500	30

```
|   {
|     empno: 7654,
|     ename: 'MARTIN',
|     job: 'SALESMAN',
|     mgr: 7698,
|     hiredate: new Date('1993-09-28'),
|     sal: 1250,
|     comm: 1400,
|     deptno: 30
|   },
|   {
|     empno: 7844,
|     ename: 'TURNER',
|     job: 'SALESMAN',
|     mgr: 7698,
|     hiredate: new Date('1981-09-08'),
|     sal: 1500,
|     comm: 0,
|     deptno: 30
|   },
|   {
|     empno: 7521,
|     ename: 'WARD',
|     job: 'SALESMAN',
|     mgr: 7698,
|     hiredate: new Date('1994-02-22'),
|     sal: 1250,
|     comm: 500,
|     deptno: 30
|   }
| ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69bcaad27205df8fc48563b8'),
    '1': ObjectId('69bcaad27205df8fc48563b9'),
    '2': ObjectId('69bcaad27205df8fc48563ba'),
    '3': ObjectId('69bcaad27205df8fc48563bb'),
    '4': ObjectId('69bcaad27205df8fc48563bc'),
    '5': ObjectId('69bcaad27205df8fc48563bd')
  }
}
test>
```

Figure 5: Inserted Data from Given Table

1.3. Exercise 2.4

1.3.1. Update the name of department 40 to: COMPUTING

```
test> db.dept.updateOne(
|   { "_id": ObjectId(40) },
|   { "$set": { dname: "ACCOUNTING" }},
| );
|
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
test>
```

Figure 6: Update Dept to ACCOUNTING

1.3.2. Update the salary of employee number 7788 in department 20 to 3500

```
}
test> db.emp.updateOne(
|   { "_id": ObjectId(7788), "deptno": 20  },
|   { "$set": { sal: 3500 }},
| );
|
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
test>
```

Figure 7: Update the salary of emp